

Infective Endocarditis (IE) - Summary

Key Symptoms: General systemic infection symptoms: fever, malaise, pain, night sweats, weight loss, etc.

- **IE-specific:** splinter hemorrhaging, Janeway lesions, Osler nodes, Roth spots, etc.

Most common causing pathogens: G+ve > G-ve

- **G positive bacteria** = Staphylococci (40%) > Streptococci (20%) > Enterococci (10%)
- **G-negative bacteria** = HACEK organisms (1-2%)
- **Others:** Difficult-to-grow pathogens (often culture negative): e.g., *C. burnetii*, *Bartonella* spp., *Chlamydia* (8%) and rarely fungal (*Candida* spp.)

Diagnosis:

- **Definite IE:** Pathological criteria + clinical criteria (2 major **OR** 1 major + 3 minor **OR** 5 minor)
- **Possible IE:** 1 major + 1 minor **OR** 3 minor
- ❖ **Major criteria:** Positive blood cultures, evidence of endocardial involvement, new valvular regurgitation, positive echocardiogram.
- ❖ **Minor criteria:** Predisposition, fever $\geq 38^{\circ}\text{C}$, vascular phenomena, immunologic phenomena, microbiological (positive culture but does not meet major criteria, see slide 13), no ECHO criteria

Non-Pharmacological Treatment:

- Surgery (for cultures and to remove vegetation if required)
 - *Brucella* spp. and *Candida* spp. nearly always require surgery

Pharmacological Treatment:

- **Staphylococci:** 6 weeks Tx (2 weeks in MSSA if right-side & uncomplicated)
 - **Native valve:** cloxacillin/cefazolin
 - **Prosthetic valve:** (cloxacillin/cefazolin + rifampin) + gentamicin only for first 2 weeks
 - **If MRSA:** vancomycin in place of cloxacillin/cefazolin in both valves
- **Streptococci:** 4-6 weeks Tx (depends on valve & MIC, see slide 22-24)
 - Pen G or ceftriaxone +/- gentamicin
 - **Beta-lactam allergy:** vancomycin (target trough: 10-15 mcg/mL)
- **Enterococci:** 4-6 weeks Tx
 - Pen G/ampicillin + gentamicin
 - **If CrCl <50mL/min:** ampicillin + ceftriaxone
 - **Beta-lactam allergy:** vancomycin (target trough: 10-15 mcg/mL) + gentamicin
- **HACEK group:** 4 weeks Tx if native valve and 6 weeks if prosthetic valve
 - Ceftriaxone
 - **Beta-lactam allergy:** fluoroquinolone and consult ID specialist
 - Avoid ampicillin/penicillin (resistance) and gentamicin (nephrotoxicity)

Monitoring Parameters: daily monitoring of symptoms/clinical signs until normalized

Additional Key Points:

- Treat with bactericidal antibiotics
- Complete therapeutic drug monitoring for aminoglycosides and vancomycin to reduce adverse drug reactions (ADR) and improve efficacy
- Vancomycin infusion reaction may occur with rapid IV infusion (<1 hr) of large doses
- TDM of gentamicin/vancomycin: after 3rd dose (pre and post) then weekly, more frequent (Q2-3d) in renal function decline